Slambient Master Blueprint: The 4-Layer Hybrid Ambient Workstation (Complete Production Edition)

This master architecture document locks in all finalized design elements, polyphony budgets, generative logic, synesthetic color-mapping UI themes, UI layout specifications, production-level reliability features (state persistence, brickwall limiting, MIDI clock sync), preset management, and plug-and-play instant generative modes for the custom **Slambient** (Sloop-FM1) firmware project.

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1. Aesthetic, Sonic Goals & Maestro Inspirations

- **The Vibe:** Warm, nostalgic, cinematic "Boards of Canada" ambient aesthetic with organic drift, emotional depth, and zero digital popping/clipping.
- **Analog Imperfections:** Built-in tape flutter / wow-and-flutter modulation running across the master output chain.
- **Dynamic Breathing & Dynamic Ducking:** Designed with natural pauses, velvet-smooth attacks, and soft rolling envelopes so the "wall of sound" blooms and rests organically. Subtle dynamic ducking (-3 dB to -6 dB) is automatically applied to the sub-drone and shimmer arpeggio during active lead phrases, maximizing harmonic clarity and space without adding extra voice burden.
- **Brian Eno Engine (Generative Phase-Locked Ecosystems & Coprime Architecture):**
    - **Coprime Multi-Layer Clock & Clock Dividers:** Multi-layer step cycles utilize extended coprime lengths (7, 11, 13, 17, 19, and 23 steps) combined with layer clock dividers (e.g., Sub-Drone steps advance every 4 bars / 64 steps, Chords advance every 2 bars / 32 steps). While raw 7-, 11-, and 13-step coprime cycles repeat every 1,001 steps (~8 minutes at ambient tempos), combining clock dividers with extended coprimes pushes structural loop periods to many hours.
    - **High-Entropy Phase Evolution, Weighted Random Walk & "Weather" Macro Layer:** High-entropy phase evolution with low repetition probability across multi-hour sessions is achieved by combining coprime multi-layer step cycles, layer clock dividers, and a weighted random walk with tonic/dominant gravitational pull (where interval probabilities naturally bound discordance to a 10% tension rule statistical target). A 20–40 minute ultra-slow drift "Weather" macro layer modulates global density and harmonic behavior over continuous multi-hour sessions.
- **Hiroshi Yoshimura Engine (Kankyo Ongaku Fluidity):**
    - **Environmental Micro-Glides (Thermal Portamento):** Fluid, variable-speed portamento curves that make harmonic shifts bend and "melt" naturally like water currents rather than stepping abruptly.

    - **Acoustic Spatial Diffusion:** Algorithmic feedback and stereo expansion scaling dynamically with drone spectral weight to simulate physical architectural space.
- **Four Tet Textures:** Acoustic-to-pad hybrid transient envelopes pairing woody plucks with lush washes.
- **Aphex Tuning & Micro-Timing:** Alternative micro-tunings, sub-millisecond event offsets within 64-sample audio blocks (~1.45 ms at 44.1 kHz) for sub-millisecond micro-timing humanization, and micro-timing swing for organic human feel.

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2. The 4-Layer Slambient Architecture

To maximize the FM-1 hardware without cluttering the frequency spectrum, the system is architected into three decoupled subsystems running off a unified master clock, while the sound engine is divided into four distinct, concurrent layers:

- **Subsystem 1: Generative Engine:** Manages 6-prime clock dividers, scale math, weighted random walk, mood presets, and seeded event scheduling.
- **Subsystem 2: Existing Synth & Sequencer Engine:** Handles track routing, note event execution, voice allocation, amplitude envelopes, and live 27-key silicone keyboard pass-through.
- **Subsystem 3: Optional Audio DSP & UI:** Manages master bus tape effects (wow/flutter), provisional reverb, lookahead peak limiting, 24 FPS strip-rendered display output, preset management, and circular sector flash persistence.

1. **Layer 1: Euclidean Rhythms (The Pulse)**
    - Generates organic, syncopated rhythmic percussion and micro-accents.
    - Uses ultra-short percussive envelopes that decay instantly so they never steal voices from the pads.
2. **Layer 2: Ambient Chords & Lead Engine (The Harmony & Voice)**
    - **Weighted Random Walk & Tonic Gravity:** Discards rigid loops in favor of a weighted random walk driven by tonic/dominant gravitational pull and interval probability distributions across scale degrees (Dorian/Lydian modal shifts), anchoring pitch selection to tonal centers.
    - **Call-and-Response Interplay:** Meandering lead phrases trigger *first* with clear presence, followed by a breathing pause and a responsive pad swell anchored to that musical space.
    - **Controlled Discordance & 10% Tension Rule Target:** The 10% tension rule is defined as a long-run statistical target with tension defined as non-diatonic or passing scale degrees outside the primary consonance set. Tightly bounded interval step probabilities ensure that tension resolves harmonically back to root stability without artificial thresholding.
3. **Layer 3: Micro-Arpeggio Generator (The Shimmer)**
    - An independent, fast-moving melodic layer that trickles notes in a randomized, non-repeating high-mid cascade.

    - Shaped with delicate velocities to sound like twinkling mist or falling water over the pad floor.
4. **Layer 4: Living Sub-Drone Bed (The Tectonic Floor)**
    - A continuous, slow-evolving spectral bed that morphs its harmonic balance over minutes rather than seconds.
    - Tracks the active root anchor and slowly swells sub-bass overtones to keep emotional tension grounded.

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3. Plug-and-Play Instant Ambient Mode ("Instant Slambient")

- **Zero-Thought Operation:** A dedicated startup mode allowing you to simply turn on the FM-1, press play, and let the device fully build out a complete, self-sustaining ambient world instantly.
- **Mood Presets:** Pre-baked macro profiles (e.g., *Nostalgic Drift, Glacial Morning, Hauntological Tape*) that automatically configure scale parameters, drone weight, Euclidean density, and arpeggio speed depending on the desired mood.

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4. Polyphony Management, Audio Safety & Memory Constraints

- **Generative Co-Processor Architecture:** Slambient operates as an internal Generative Event/MIDI Co-Processor triggering notes, CCs, and envelopes into the stock SLOOP/Felucca synthesis and drum engine (Tracks 1–4) rather than attempting to synthesize PCM waveforms from scratch inside the high-priority audio ISR. The pre-computed `SINE_LUT[1024]` lookup table is used solely for low-frequency control signals (LFOs), tape wow/flutter delay modulation, thermal portamento pitch curves, and synesthetic UI polygon rendering.
- **DSP Asymmetry Principle (Drum vs. Synth Voice Cost):** Drum voices and synth voices run on completely separate DSP pipelines and are not 1:1 interchangeable in CPU cycles. Drum voices are lightweight one-shot triggers with short decay (50–200 ms), whereas synth voices (especially 6-op FM6 and 4-pole resonant VA) require intensive per-sample DSP calculations.
- **Authoritative Voice Allocation Table (8 Synth Voices + 4 Drum Voices):** The polyphony budget is strictly bound across an authoritative allocation table to guarantee CPU safety and zero voice starvation:
    - **Voice 0 (Track 1):** Sub-Drone (Layer 4 continuous mono bed).
    - **Voices 1–2 (Track 2):** Chord Pad Root & 5th (Layer 2 polyphonic pad floor).
    - **Voice 3 (Track 2):** Shared Pad/Lead Voice (borrowed by Lead during active melodic phrases, returning seamlessly to Pad as 9th/triad extension during breathing pauses).
    - **Voice 4 (Track 3):** Shimmer Arpeggio (Layer 3 mono cascade).

    - **Voices 5–7 (Track 2):** Reserved Live Performance Headroom (3 synth voices strictly reserved for real-time play on the 27 silicone keys).
    - **Track 4 (Isolated Drum Engine):** 4 dedicated, isolated drum voices (micro-accents, kick, click, shaker).
    - **Voice Summary:** Total synth voices = 8. Active generative voices = 5 (Drone 1 + Pad/Lead 3 + Arp 1). Live performance headroom = 3. Isolated drum voices = 4.
    - **Cap Drum Voices at 4 (Layer 1):** Capped at 4 drum voices (reduced from stock 6). Euclidean ambient percussion only fires 2–3 simultaneous micro-accents (kick, woody click, shaker), so capping at 4 eliminates unnecessary CPU spikes during dense polyrhythmic passages. Includes global Percussion Filter Damp and transient softening to gently integrate percussive textures.
    - **Provisional Master FX & Lean Half-Rate Reverb:** Master bus effects (wow/flutter delay modulation, tanh soft-saturation, lookahead peak limiter, and half-rate diffusion reverb) are designated as provisional pending Phase 2 source code audit. Post-saturation sample-peak limiting with soft saturation prevents digital clipping during dense voice swells. The optional diffusion reverb network runs at half sample rate (22.05 kHz) with a low-pass darker tail to fit within audio memory constraints (~15–20 KB). To prevent hung MIDI voices, an automatic Stuck-Note Watchdog enforces layer-specific duration limits (Lead/Arp/Drums max 15s; Chords max 30s with 50 ms crossfade; Sub-Drone exempt). Critical audio operations (note-offs, voice releases, panic commands) bypass the UI display FIFO entirely and execute directly via atomic/guaranteed handlers.
    - **Explicit Voice-Stealing Hierarchy & Amplitude Envelopes:** Voice preemption follows a strict priority hierarchy: `Live Keys (PRIORITY_CRITICAL)` > `Lead Melodic Voice` > `Chord Pad Voicings` > `Shimmer Arp` > `Sub-Drone (LOWEST)`. Voice transitions are controlled by amplitude envelopes rather than zero-crossing detectors: when a voice is preempted, it executes a 10 ms cosine amplitude ramp-down to eliminate preemption clicks during voice reallocation, while chord and modal transitions use a 50–100 ms exponential amplitude release.
    - **Preserve the Reliable 8-Synth Pool:** Rather than risking RAM overflow or ISR starvation by pushing to 9+ synth voices, keep 8 synth voices and resolve harmonic density via dynamic Voice Borrowing (lead voice borrows Voice 3 of the chord pad during phrases, which swells back into a full triad during breathing pauses).
    - **Authoritative Summary:** 8 Synth Voices + 4 Drum Voices = 12 total managed voices across Subsystems 1 & 2.
- **Zero-Crossing Audio Safety & Interpolation:** Smooth voice release windows (50–100 ms exponential fade) are enforced during modal shifts and auto-latch generative hand-offs to avoid zero-crossing clicks. Delay-line read pointer modulations for wow/flutter and thermal portamento utilize linear/hermite interpolation to completely prevent zipper noise.

- **Provisional Master Peak Limiter & Soft-Saturation:** Master bus processing combines tanh soft-saturation with a true sample-peak ceiling placed directly post-saturation to catch inter-sample overshoots, preserving warm tape grit while protecting output levels.
- **RAM & Display Strip-Rendering Architecture:** Static `.data+.bss` RAM has a strict ~98 KB ceiling with only ~5.5 KB of headroom remaining. Allocating a full 240x240 16-bit display framebuffer (~115 KB) exceeds available RAM, and full-screen 240x240 16-bit rendering over SPI takes 23–46 ms, exceeding the 41 ms (24 FPS) rendering budget. The UI avoids full-screen per-pixel watercolor blending, using instead coarse blended color zones and strictly bounded dirty rectangles (orbital arcs, center text, and dials). The 4–8 KB canvas tile is allocated once at boot from the dynamic audio heap, marked non-cacheable/DMA-capable, with zero runtime allocation or freeing to preserve the 5.5 KB BSS RAM. Sub-millisecond event scheduling handles event offsets within 64-sample audio blocks (~1.45 ms at 44.1 kHz) to preserve micro-timing. Furthermore, all generative and modulation math uses 32-bit fixed-point integer arithmetic referencing the `SINE_LUT[1024]` table to prevent floating-point pipeline stalls inside audio interrupts.

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5. Hardware Integration & Real-World Limitations

- **Live Pass-Through:** Actively playing the device's 27 silicone keys bypasses generation and sends your notes straight to the synth engine.
- **Leaky Pitch-Class Histogram & Mode Inference:** As manual performance notes are played on the 27 silicone keys, a leaky pitch-class histogram analyzes note distribution in real-time, inferring both the root anchor and scale mode (e.g., Dorian vs. Lydian) from manual play.
- **Auto-Latch Seamless Transition:** When no keys are touched for a brief threshold (e.g., 3 seconds), an internal flag triggers a harmonic hand-off. The engine captures the inferred root and modal scale, seamlessly reassigning them as the new generative anchor for wandering bass, lead, and pads.
- **Bluetooth BLE Hardware Disable:** In custom SLOOP/Felucca firmwares, the Bluetooth BLE radio is intentionally powered off and kept disabled to eliminate RF interference bleeding into the DAC and prevent BLE interrupt latency. The projected 60 KB RAM savings from BLE disablement is unverified pending full linker map analysis in Step 2. All MIDI communication is handled exclusively via class-compliant USB-C MIDI and the 3.5mm TRS hardware MIDI IN jack.
- **Flash Journaling Ring Layout & Experimental OTA Rollback:** Flash persistence uses a log-structured ring buffer across a dedicated 4 KB sector with a compact 24-byte record layout consisting of sequence numbers, active world seed, macro states, and rolling CRC32 checksum. Flash writes execute every few minutes to pre-erased sector ring entries, on transport STOP, manual SAVE, or upon crossing a 5% low-battery threshold. Sector erases occur strictly during Eno silence pauses or transport STOP. Hardware dual-bank OTA rollback capabilities are flagged as experimental until physically tested and verified on hardware.

- **Generative "World" Seeds:** Entire generative session states are represented as a compact 32-bit xorshift PRNG seed combined with macro parameter settings. The full 32-bit xorshift seed is formatted as an 8-character hex code (e.g., `#A3F2-B8D4`), with a 4-character truncated shorthand (e.g., `#D7A4`) used strictly as a display-only shorthand on compact UI views; recall and deterministic session reproduction strictly require the full 32-bit / 8-character hex code.
- **Physical Hardware Knobs & External MIDI Sync:** Dedicated physical knobs provide direct tactile control (Knob 1: Mutation Density, Knob 2: Drone Weight, Knob 3: Shimmer Speed, Knob 4: Spatial Diffusion/Reverb). The SELECT macro knob ("Weather") smoothly interpolates overall density, BPM (40–75 BPM), drone overtone weight, and thermal portamento drift. When an external MIDI clock signal is detected, the "Weather" macro's internal BPM modulation automatically decouples—keeping the external clock authoritative while continuing to modulate density, drone weight, and portamento.
- **Layer 1 Rhythmic Evolution:** To ensure the 7-step Euclidean drum pattern continuously evolves without feeling repetitive, Layer 1 incorporates per-bar probability gating (chance), dynamic fill muting, and subtle pattern rotations every 2 or 4 bars.
- **Power-of-Two Drop-on-Full FIFO Decoupling:** The lossy UI display FIFO buffer is sized to 32 entries (power-of-two) and drops incoming visual events on full, ensuring the display thread never stalls the audio ISR. Critical audio event paths (note-offs, voice releases, panic commands) bypass the lossy UI FIFO entirely and execute directly via guaranteed atomic event handlers.
- **Audio I/O Specification & USB Audio Direction:** The 3.5mm analog jack operates as stereo audio output only (with automatic internal speaker mute). Stock SLOOP USB Audio operates strictly as a USB Audio Device Class output to host (allowing a host computer to record the FM-1 master output). External audio input into the FM-1 from a host is unverified and flagged as an unconfirmed dependency; external-input DSP processing will not be planned until verified in source code.

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6. Modular C Firmware Structure (`slambient-fm1`)

The codebase is cleanly modularized into three decoupled subsystems, aligning with the implementation plan across `slambient.h / slambient.c` (Subsystem 1: Generative Engine), `audio.c / seq.c / voice.c` (Subsystem 2: Synth & Sequencer Engine), and `slambient_fx.c / ui_studio.c / presets.c` (Subsystem 3: Optional Audio DSP, UI & Flash Persistence):

- **`slambient.h / slambient.c (Subsystem 1):`** Core Generative Engine managing 6-prime clock dividers, scale math, stateful random-walk stepping, modal drift rules, micro-tunings, thermal portamento, mood presets, and lock-free FIFO ring buffer queuing to Subsystem 3.

- **`audio.c / seq.c / voice.c (Subsystem 2):`** Existing Synth & Sequencer Engine handling track routing, note event execution, 8-synth/4-drum voice allocation table, amplitude envelopes, and live 27-key silicone keyboard pass-through.
- **`slambient_fx.c / ui_studio.c (Subsystem 3):`** Optional Audio DSP and Display Engine managing provisional master bus tape wow/flutter, soft-saturation, lookahead peak limiting, and non-blocking 24 FPS strip-rendered display updates.
- **`presets.c / .h`**: Manages onboard storage, recall of custom ambient user slots, instant mood profiles, and Auto-State Snapshots with pre-erased flash journal sector management.

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7. Slambient Hardware UI, Synesthetic Engine & TFT Visualizer

- **Boot Screen:** Custom minimalist "Slambient" startup animation with a breathing analog wave pattern.
- **Synesthetic Color Mapping Engine:** Pitch classes, scale modes, and chord intervals dynamically correspond to specific color coordinates (e.g., deep earthy roots mapping to warm amber/ochre, ethereal thirds/fifths shifting into twilight indigos, and sparkling arpeggios adding vibrant moss greens or celestial cyans).
- **Harmonic Color Blending:** As notes overlap across the 4 layers, the TFT display real-time blends these chromatic values together across the UI elements, creating an evolving, living watercolor effect that mirrors the audio.
- **TFT Center Display:** Bold, clear text showing the active root note and chord voicing (e.g., `D : WARM_TRIAD`) rendered in the primary synesthetic color of the chord.
- **Harmonic Orbit & Layer Arcs:** A morphing organic center polygon accompanied by four concentric vector arcs representing the 4 prime-number layers as they slowly drift and phase-lock over time.
- **Battery Conservation, Auto-Dim & Diagnostic Instruments:** Includes a 60-second display auto-dimmer that drops backlight brightness to 30% during generative idle, extending battery runtime to 10–12 hours. A development overlay provides real-time diagnostic instruments including an audio buffer underrun counter and a CPU load percentage meter.
- **TFT Display Architecture & Refresh Rate:** Visual refresh rates are locked to a cinematic 24 FPS (filmic / classic hand-drawn animation frame rate) to evoke the vintage Boards of Canada tape aesthetic while capping CPU overhead and SPI bus saturation. Screen updates utilize non-blocking DMA framebuffer transfers and dirty-rectangle rendering for the harmonic orbit polygon, concentric arcs, and watercolor blending to prevent audio buffer underruns.

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8. Hardware Verification & Pre-Flight Checklist

- **Local Build Toolchain:** Verified build pipeline running the JieLi `pi32v2` toolchain (located at `~/.jieli/toolchain`) via Docker Desktop on macOS.

- **Target Binary Package:** Compiles to `build/felucca.fwsc` with package ID `FM-1_900`.
- **Flashing Procedure:** Flashed via USB-C using WebMIDI in Chrome/Edge at fm1.designburgapps.com, featuring instant single-click rollback to stock V15 firmware.
- **Desktop Simulator First & Golden-Seed Regression Tests (Step Zero):** Prior to deploying to physical hardware, compile `generator.c` and `sequencer.c` as a standalone macOS command-line executable outputting MIDI directly to a DAW or via RtMidi. Simulator determinism relies on an abstracted virtual clock with recorded timeline inputs, eliminating unseeded rand() calls or real-time jitter. Automated golden-seed CLI tests verify that identical world seeds produce byte-identical event streams across multi-hour runs before flashing firmware.

9. 8-Step Engineering Roadmap & Implementation Plan

This section outlines the authoritative 8-step build order for engineering, auditing, integrating, and verifying the Slambient firmware on physical hardware.

Step 1: Establish Baseline & Clean Build Verification

- Compile stock SLOOP firmware binary using the JieLi toolchain in Docker. Verify clean compilation, inspect baseline flash and BSS RAM usage, establish WebMIDI flashing pipeline on test unit, and verify 100% stock functionality.

Step 2: Source Code & Hook Audit

- Inspect `audio.c`, `fx.c`, `seq.c`, and `voice.c`. Audit post-mix audio bus tap points, sample-accurate sub-block event offsets, voice allocation preemption hooks, USB audio streaming direction, and linker map memory allocations (verifying BSS headroom and BLE disablement RAM impact).

Step 3: Desktop Simulator Verification (macOS Native CLI)

- Compile Subsystem 1 (generator.c, sequencer.c) natively on macOS with RtMidi output. Run automated statistical regression suite across 1,000,000 steps to verify 10% tension rule statistical target, zero voice leaks across the 8-voice table, deterministic golden-seed output, and high-entropy window hashing.

Step 4: Single-Layer Hardware Integration

- Integrate Subsystem 1 into embedded stock codebase driving Layer 2 (Pad/Lead) only. Verify event trigger reliability, 27-key silicone keyboard live pass-through on Track 2, and smooth exponential amplitude releases on modal transitions.

Step 5: Multi-Layer Expansion & 8-Voice Accounting Stress Test

- Enable all 4 generative layers (Drone, Chords/Lead, Arp, Drums). Execute rigorous stress testing of the authoritative voice allocation table (Voices 0–4 generative, Voices 5–7 live keys, Track 4 drums). Verify Voice 3 dynamic borrowing, Stuck-Note Watchdog timeouts, and zero voice starvation during heavy live play.

Step 6: Master DSP & Audio Tap Insertion

- Insert Subsystem 3 master audio processing (wow/flutter delay modulation, tanh soft-saturation, lookahead peak limiter, half-rate diffusion reverb) into master audio buffer loop. Measure CPU load, buffer underruns, and DSP latency on hardware.

Step 7: 24 FPS Strip-Rendered UI & Flash Persistence

- Connect Subsystem 1 event FIFO to Subsystem 3 display thread. Implement 24 FPS non-blocking frame limiter, DMA dirty-rectangle rendering on heap canvas tile, and 24-byte circular sector flash journaling with low-battery auto-save.

Step 8: Physical Hardware Verification & Long Soak Testing

- Perform continuous 24-hour burn-in soak test on physical hardware. Monitor thermal stability, audio buffer underrun counters, battery power drop-off behavior, pre-erased flash journaling integrity, and USB MIDI log parity against simulator golden seed outputs.